

Inventory Management System (IMS) Database Design

Author: Muhammad Bilal

bilalmubeen99@gmail.com

Abstract

This project presents the comprehensive design and implementation of a robust Inventory Management System (IMS) database aimed at optimizing stock control, supplier relations, and transaction tracking. To ensure maximum data integrity and minimize redundancy, the relational schema was rigorously normalized to Boyce-Codd Normal Form (BCNF). The system architecture encompasses seven core entities, including Products, Categories, Suppliers, Customers, Stock Transactions, Returns, and Purchase Orders, organized through strategically defined one-to-many and one-to-one relationships.

Implementation was carried out using SQL, featuring advanced database objects such as multi-table joins for complex data retrieval, nested sub-queries for analytical insights, and specialized Views for real-time monitoring of stock levels and return summaries. Furthermore, the system incorporates Stored Procedures with built-in transaction management and rollback capabilities to guarantee atomicity during critical operations like stock updates and price adjustments. The resulting database provides a scalable and efficient foundation for enterprise-level resource planning and data-driven decision-making.

Keywords

Database Management Systems (DBMS), Relational Database Design, Boyce-Codd Normal Form (BCNF), Data Normalization, Inventory Control, SQL, Stored Procedures, Data Integrity, Entity-Relationship Diagram (ERD), Transaction Management.

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1. Introduction

An Inventory Management System (IMS) is a crucial tool for businesses to efficiently track, manage, and control their inventory. It ensures proper stock levels, reduces waste, streamlines operations, and enhances decision-making. This document outlines the design of a database for an IMS, adhering to the principles of normalization, particularly Boyce-Codd Normal Form (BCNF).

2. Objectives of the IMS Database

- 2.1. Efficient Stock Management:** Track inventory levels, sales, purchases, and returns.
- 2.2. Supplier Management:** Maintain detailed records of suppliers and streamline purchase orders.
- 2.3. Customer Tracking:** Manage customer information and order history.
- 2.4. Data Integrity:** Ensure accurate, consistent, and normalized data.
- 2.5. Scalability:** Support future business growth and additional features.

3. Deliverables

- A relational database schema designed in BCNF.
- Detailed attributes for each entity.
- Logical relationships between tables.
- Description of normalization applied.

4. Entities and Their Attributes

The database will consist of the following entities:

4.1. Products

Attributes:

- Product_ID (Primary Key)
- Product_Name
- Category_ID (Foreign Key)
- SKU (Stock Keeping Unit)
- Price

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- Stock_Level
- Description

4.2. Categories

Attributes:

- Category_ID (Primary Key)
- Category_Name
- Description

4.3. Suppliers

Attributes:

- Supplier_ID (Primary Key)
- Supplier_Name
- Contact_Name
- Phone_Number
- Email
- Address

4.4. Stock Transactions

Attributes:

- Transaction_ID (Primary Key)
- Product_ID (Foreign Key)
- Transaction_Type (e.g., Sale, Purchase, Adjustment)
- Quantity
- Transaction_Date
- Remarks

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4.5. Returns

Attributes:

- Return_ID (Primary Key)
- Sale_Transaction_ID (Foreign Key)
- Product_ID (Foreign Key)
- Quantity
- Reason
- Return_Date

4.6. Customers

Attributes:

- Customer_ID (Primary Key)
- Customer_Name
- Phone_Number
- Email
- Address

4.7. Purchase Orders

Attributes:

- Order_ID (Primary Key)
- Supplier_ID (Foreign Key)
- Order_Date
- Expected_Delivery_Date
- Status (e.g., Pending, Completed)
- Total_Amount

5. Database Normalization and BCNF

The following normalization steps ensure the database is in BCNF:

1. **Eliminate Repeating Groups:** Each attribute contains atomic values.
2. **Eliminate Partial Dependencies:** All non-prime attributes depend only on the primary key.
3. **Eliminate Transitive Dependencies:** Non-prime attributes depend directly on the primary key.
4. **BCNF Compliance:** Every determinant is a candidate key.

Example of BCNF Compliance

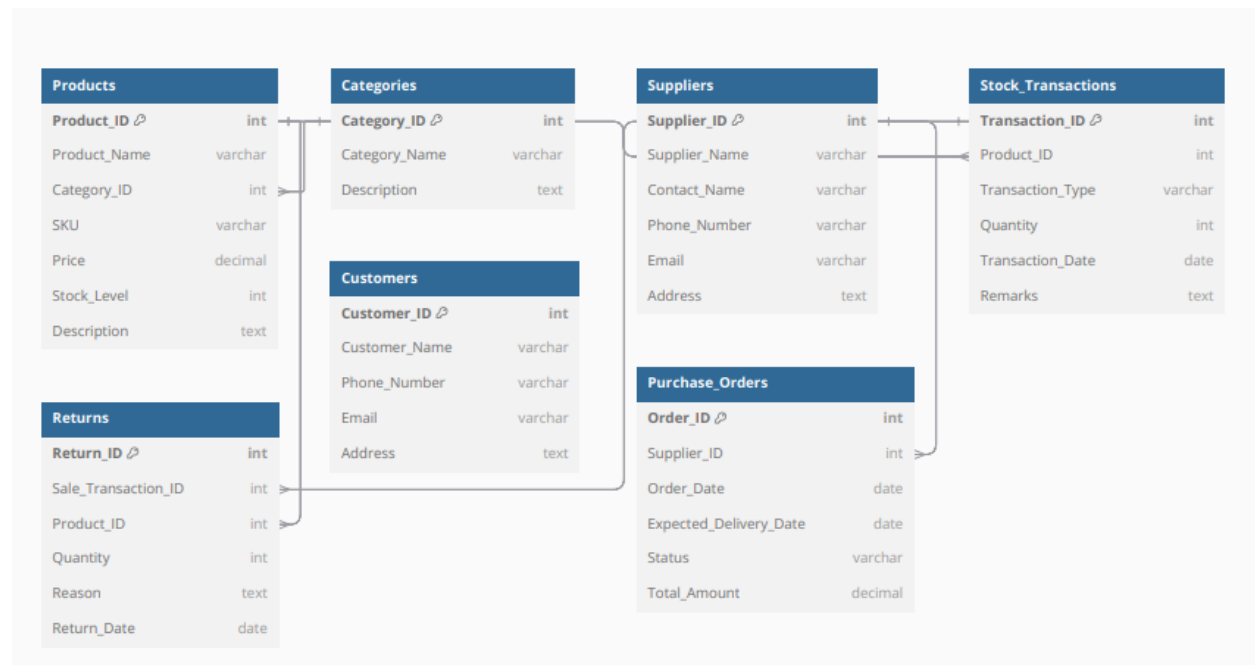
For the Products table:

- **Primary Key:** Product_ID
- Non-prime attributes (Product_Name, Category_ID, SKU, Price, Stock_Level, Description) depend solely on Product_ID.
- No transitive or partial dependencies exist.

6. Relationships Between Tables

- **Products → Categories:** One-to-Many (Each product belongs to a single category).
- **Products → Stock Transactions:** One-to-Many (A product can have multiple stock transactions).
- **Suppliers → Purchase Orders:** One-to-Many (A supplier can have multiple purchase orders).
- **Customers → Returns:** One-to-Many (A customer can make multiple returns).
- **Stock Transactions → Returns:** One-to-One (Each return is tied to a specific transaction).

7. ERD Diagram



8. Analysis

The database has been normalized to BCNF by ensuring:

- No repeating groups (atomic attributes).
- Non-prime attributes depend only on the primary key.
- All determinants are candidate keys.
- The relationships between entities (e.g., Products, Categories, Suppliers) follow one-to-many and one-to-one associations as necessary.

9. Results

The IMS database design adheres to BCNF principles, ensuring accurate data storage, minimized redundancy, and efficient relationships between tables.

10. Discussion on Results

By normalizing the database to BCNF, data consistency and integrity are enhanced. Redundant information is minimized, and all entities are directly related to the primary key without transitive dependencies.

11. Conclusion

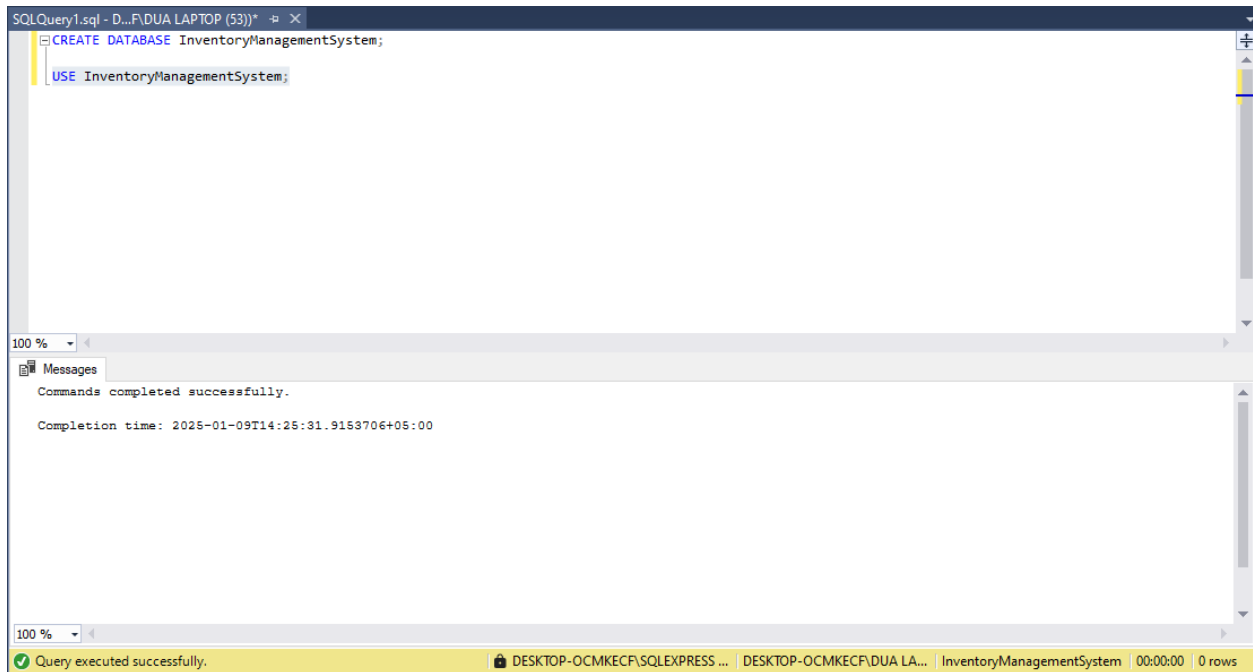
This database design ensures data integrity, scalability, and efficient inventory management. The structure adheres to BCNF principles, reducing redundancy and improving data consistency. The IMS database provides a robust foundation for tracking inventory, managing suppliers and customers, and handling transactions and returns effectively.

12. Reference

- Database Normalization Concepts and Principles.
- BCNF and Database Design Documentation.

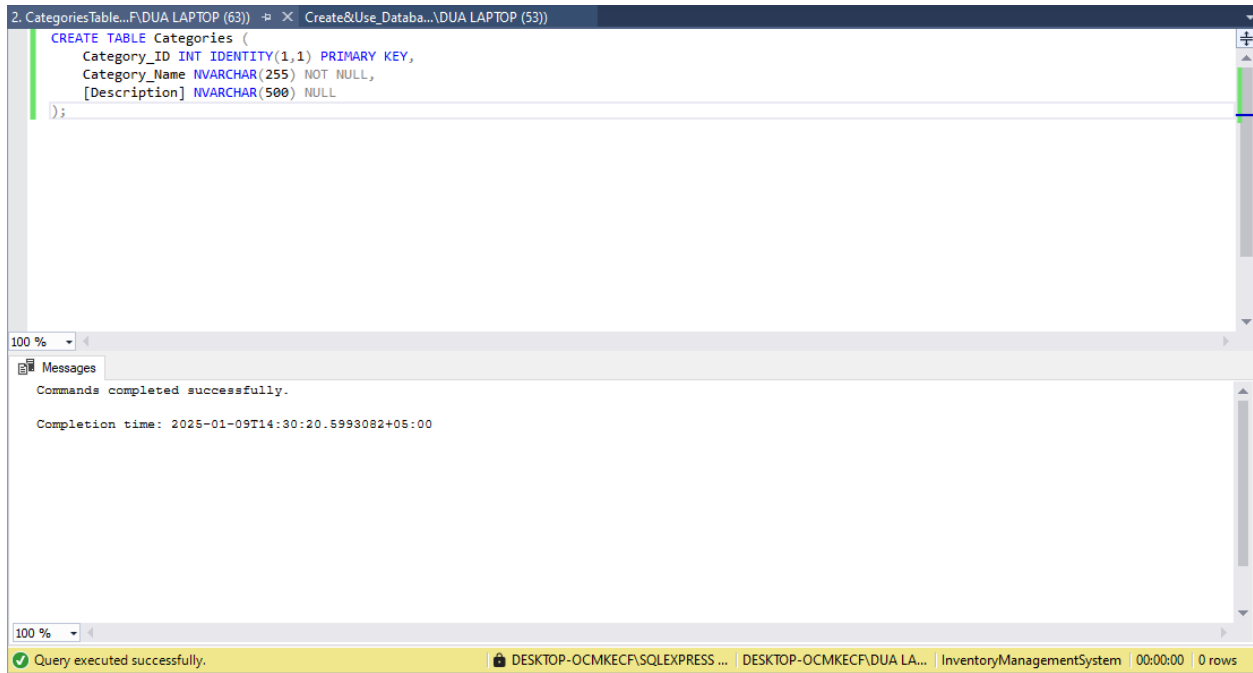
13. Screen Shots

13.1. Creating Database



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13.2. Categories Table



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows the SQL script for creating the 'Categories' table. The bottom pane shows the 'Messages' window with the confirmation message 'Commands completed successfully.' and the completion time '2025-01-09T14:30:20.5993082+05:00'. The status bar at the bottom indicates 'Query executed successfully.' and '0 rows'.

```
2. CategoriesTable...F:\DUA LAPTOP (63))  Create&Use_Databa...\DUA LAPTOP (53))
CREATE TABLE Categories (
  Category_ID INT IDENTITY(1,1) PRIMARY KEY,
  Category_Name NVARCHAR(255) NOT NULL,
  [Description] NVARCHAR(500) NULL
);
```

100 %

Messages

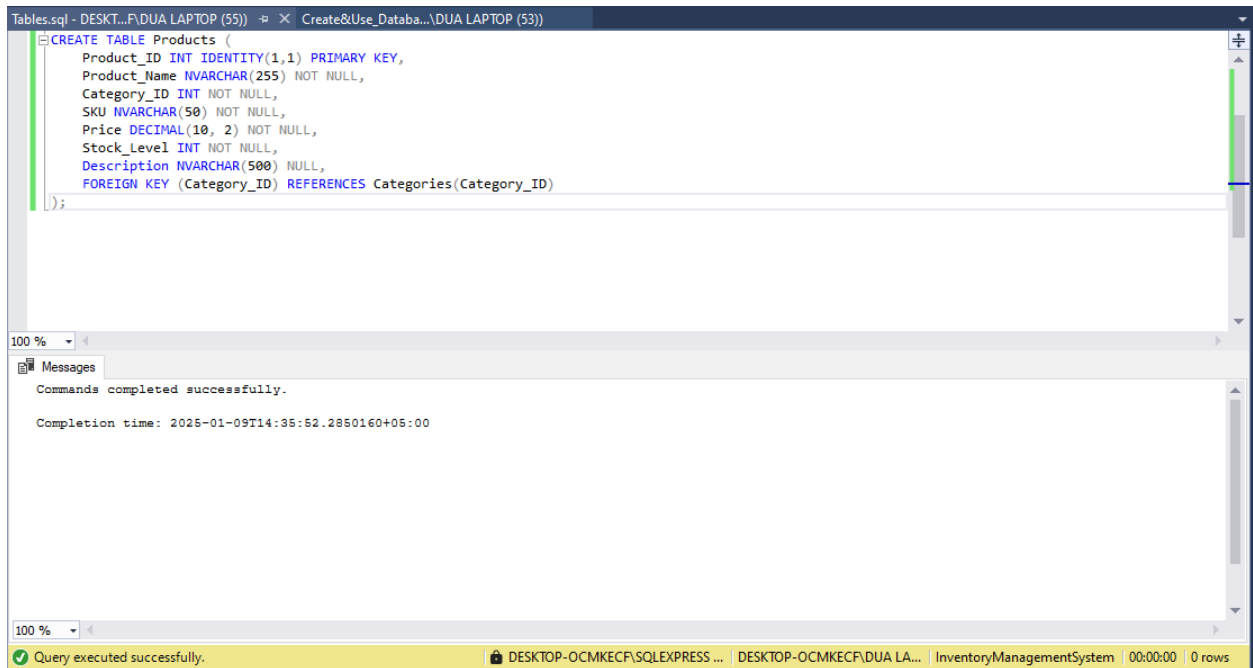
Commands completed successfully.

Completion time: 2025-01-09T14:30:20.5993082+05:00

100 %

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 0 rows

13.3. Products Table



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows the SQL script for creating the 'Products' table, including a foreign key constraint. The bottom pane shows the 'Messages' window with the confirmation message 'Commands completed successfully.' and the completion time '2025-01-09T14:35:52.2850160+05:00'. The status bar at the bottom indicates 'Query executed successfully.' and '0 rows'.

```
Tables.sql - DESKTOP-OCMKECF\SQLEXPRESS ... Create&Use_Databa...\DUA LAPTOP (53))
CREATE TABLE Products (
  Product_ID INT IDENTITY(1,1) PRIMARY KEY,
  Product_Name NVARCHAR(255) NOT NULL,
  Category_ID INT NOT NULL,
  SKU NVARCHAR(50) NOT NULL,
  Price DECIMAL(10, 2) NOT NULL,
  Stock_Level INT NOT NULL,
  Description NVARCHAR(500) NULL,
  FOREIGN KEY (Category_ID) REFERENCES Categories(Category_ID)
);
```

100 %

Messages

Commands completed successfully.

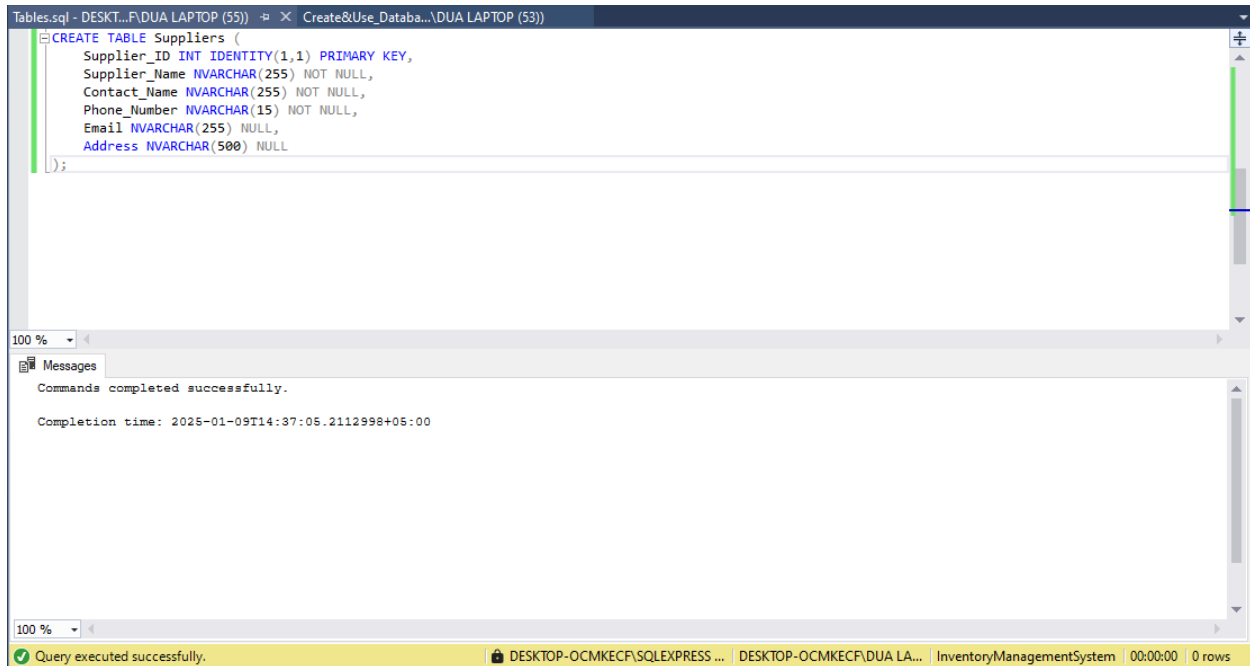
Completion time: 2025-01-09T14:35:52.2850160+05:00

100 %

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 0 rows

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13.4. Suppliers Table



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows the SQL script for creating the 'Suppliers' table. The script defines columns: Supplier_ID (INT IDENTITY(1,1) PRIMARY KEY), Supplier_Name (NVARCHAR(255) NOT NULL), Contact_Name (NVARCHAR(255) NOT NULL), Phone_Number (NVARCHAR(15) NOT NULL), Email (NVARCHAR(255) NULL), and Address (NVARCHAR(500) NULL). The bottom pane shows the 'Messages' tab with the confirmation: 'Commands completed successfully.' and 'Completion time: 2025-01-09T14:37:05.2112998+05:00'. The status bar at the bottom indicates 'Query executed successfully.' and '0 rows'.

```
CREATE TABLE Suppliers (
    Supplier_ID INT IDENTITY(1,1) PRIMARY KEY,
    Supplier_Name NVARCHAR(255) NOT NULL,
    Contact_Name NVARCHAR(255) NOT NULL,
    Phone_Number NVARCHAR(15) NOT NULL,
    Email NVARCHAR(255) NULL,
    Address NVARCHAR(500) NULL
);
```

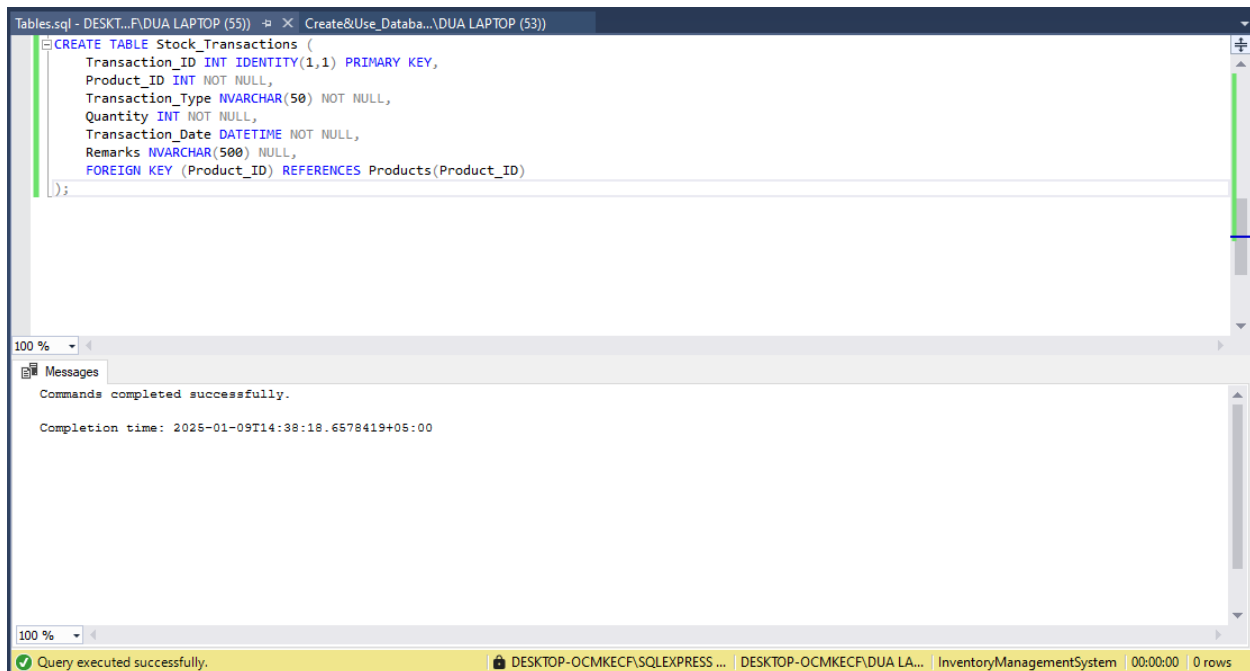
Messages

Commands completed successfully.

Completion time: 2025-01-09T14:37:05.2112998+05:00

Query executed successfully. | DESKTOP-OCMKECF\SQLEXPRESS ... | DESKTOP-OCMKECF\DUA LA... | InventoryManagementSystem | 00:00:00 | 0 rows

13.5. Stock Transactions Table



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows the SQL script for creating the 'Stock_Transactions' table. The script defines columns: Transaction_ID (INT IDENTITY(1,1) PRIMARY KEY), Product_ID (INT NOT NULL), Transaction_Type (NVARCHAR(50) NOT NULL), Quantity (INT NOT NULL), Transaction_Date (DATETIME NOT NULL), and Remarks (NVARCHAR(500) NULL). It also includes a foreign key constraint: FOREIGN KEY (Product_ID) REFERENCES Products(Product_ID). The bottom pane shows the 'Messages' tab with the confirmation: 'Commands completed successfully.' and 'Completion time: 2025-01-09T14:38:18.6578419+05:00'. The status bar at the bottom indicates 'Query executed successfully.' and '0 rows'.

```
CREATE TABLE Stock_Transactions (
    Transaction_ID INT IDENTITY(1,1) PRIMARY KEY,
    Product_ID INT NOT NULL,
    Transaction_Type NVARCHAR(50) NOT NULL,
    Quantity INT NOT NULL,
    Transaction_Date DATETIME NOT NULL,
    Remarks NVARCHAR(500) NULL,
    FOREIGN KEY (Product_ID) REFERENCES Products(Product_ID)
);
```

Messages

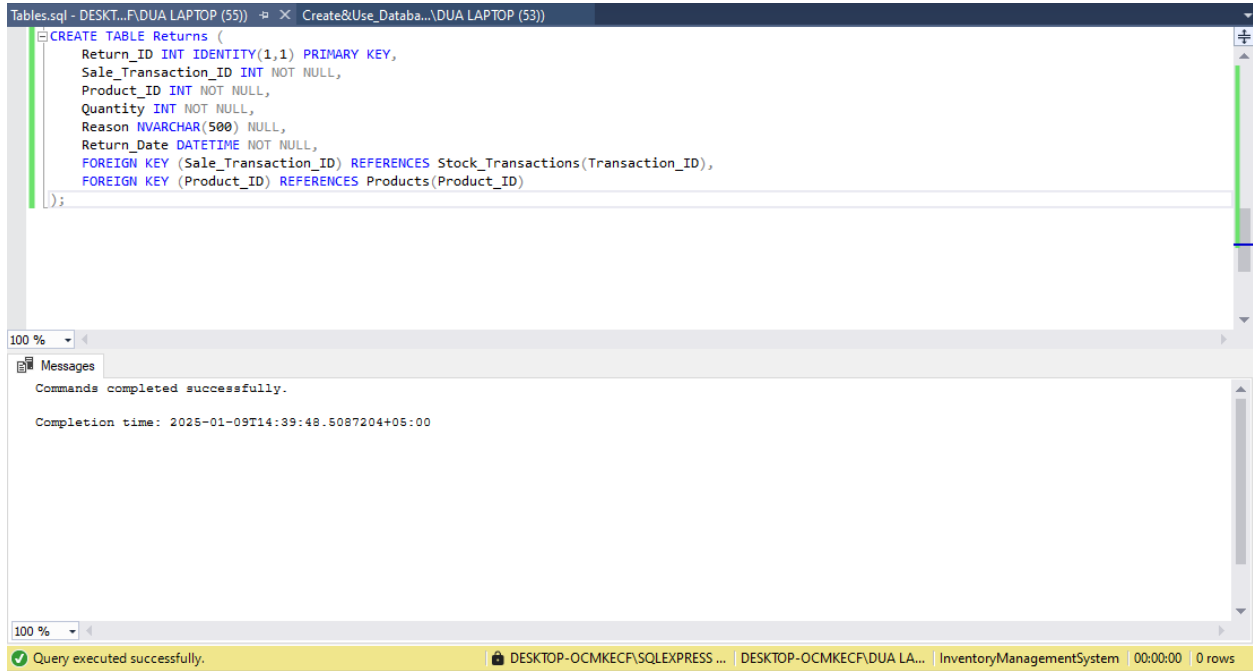
Commands completed successfully.

Completion time: 2025-01-09T14:38:18.6578419+05:00

Query executed successfully. | DESKTOP-OCMKECF\SQLEXPRESS ... | DESKTOP-OCMKECF\DUA LA... | InventoryManagementSystem | 00:00:00 | 0 rows

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13.6. Returns Table



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a SQL query window with the following code:

```
CREATE TABLE Returns (  
    Return_ID INT IDENTITY(1,1) PRIMARY KEY,  
    Sale_Transaction_ID INT NOT NULL,  
    Product_ID INT NOT NULL,  
    Quantity INT NOT NULL,  
    Reason NVARCHAR(500) NULL,  
    Return_Date DATETIME NOT NULL,  
    FOREIGN KEY (Sale_Transaction_ID) REFERENCES Stock_Transactions(Transaction_ID),  
    FOREIGN KEY (Product_ID) REFERENCES Products(Product_ID)  
);
```

The bottom pane shows the Messages tab with the following output:

```
Commands completed successfully.  
  
Completion time: 2025-01-09T14:39:48.5087204+05:00
```

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 0 rows

13.7. Customers Table



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a SQL query window with the following code:

```
CREATE TABLE Customers (  
    Customer_ID INT IDENTITY(1,1) PRIMARY KEY,  
    Customer_Name NVARCHAR(255) NOT NULL,  
    Phone_Number NVARCHAR(15) NOT NULL,  
    Email NVARCHAR(255) NULL,  
    Address NVARCHAR(500) NULL  
);
```

The bottom pane shows the Messages tab with the following output:

```
Commands completed successfully.  
  
Completion time: 2025-01-09T14:40:28.8230919+05:00
```

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 0 rows

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13.8. Purchase Order Table

```
Tables.sql - DESKTOP...F:\DUA LAPTOP (55))  Create&Use_Databa...\DUA LAPTOP (53))
CREATE TABLE Purchase_Orders (
    Order_ID INT IDENTITY(1,1) PRIMARY KEY,
    Supplier_ID INT NOT NULL,
    Order_Date DATETIME NOT NULL,
    Expected_Delivery_Date DATETIME NOT NULL,
    Status NVARCHAR(50) NOT NULL,
    Total_Amount DECIMAL(10, 2) NOT NULL,
    FOREIGN KEY (Supplier_ID) REFERENCES Suppliers(Supplier_ID)
);

Messages
Commands completed successfully.

Completion time: 2025-01-09T14:41:31.9819347+05:00

Query executed successfully.  DESKTOP-OCMKECF\SQLEXPRESS ...  DESKTOP-OCMKECF\DUA LA...  InventoryManagementSystem  00:00:00  0 rows
```

13.9. Categories Data

```
Categories Data.sql...F:\DUA LAPTOP (60))  Tables.sql - DESKTOP...F:\DUA LAPTOP (55))  Create&Use_Databa...\DUA LAPTOP (53))
CREATE TABLE Categories (
    Category_ID INT IDENTITY(1,1) PRIMARY KEY,
    Category_Name NVARCHAR(255) NOT NULL,
    Description NVARCHAR(500) NULL
);

INSERT INTO Categories (Category_Name, Description)
VALUES
    ('Electronics', 'Devices like phones, laptops, and gadgets'),
    ('Furniture', 'Chairs, tables, and other furniture items'),
    ('Clothing', 'Apparel for men and women'),
    ('Food & Beverages', 'Groceries and consumable items'),
    ('Toys', 'Toys for children of all ages'),
    ('Books', 'Various genres of books');

SELECT * FROM Categories

Results  Messages
Category_ID  Category_Name  Description
1  1  Electronics  Devices like phones, laptops, and gadgets
2  2  Furniture  Chairs, tables, and other furniture items
3  3  Clothing  Apparel for men and women
4  4  Food & Beverages  Groceries and consumable items
5  5  Toys  Toys for children of all ages
6  6  Books  Various genres of books

Query executed successfully.  DESKTOP-OCMKECF\SQLEXPRESS ...  DESKTOP-OCMKECF\DUA LA...  InventoryManagementSystem  00:00:00  6 rows
```

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13.10. Products Data

Products Data.sql...F:\DUA LAPTOP (57) Tables.sql - DESKTOP...F:\DUA LAPTOP (55) Create&Use_Databa...DUA LAPTOP (53)

```
INSERT INTO Products (Product_Name, Category_ID, SKU, Price, Stock_Level, Description)
VALUES
('Smartphone', 1, 'SP123', 499.99, 50, 'Latest model with high-end features'),
('Laptop', 1, 'LT456', 899.99, 30, 'Powerful laptop for professionals'),
('Sofa', 2, 'SF789', 299.99, 10, 'Comfortable and stylish sofa'),
('Dining Table', 2, 'DT101', 199.99, 5, 'Wooden dining table with 4 chairs'),
('T-shirt', 3, 'TS111', 19.99, 200, 'Cotton t-shirt for casual wear'),
('Jeans', 3, 'JN222', 39.99, 150, 'Denim jeans for men'),
('Pasta', 4, 'FS333', 2.99, 500, 'Pack of dry pasta'),
('Coffee', 4, 'FB444', 9.99, 150, 'Ground coffee for brewing'),
('Action Figure', 5, 'TF555', 12.99, 100, 'Toy action figure for kids'),
('Doll', 5, 'TF666', 15.99, 80, 'Doll for young children'),
('Fiction Novel', 6, 'BK777', 14.99, 120, 'Mystery fiction novel'),
('Cookbook', 6, 'BK888', 19.99, 60, 'Recipes and cooking tips');

SELECT * FROM Products
```

100 %

Results Messages

	Product_ID	Product_Name	Category_ID	SKU	Price	Stock_Level	Description
1	1	Smartphone	1	SP123	499.99	50	Latest model with high-end features
2	2	Laptop	1	LT456	899.99	30	Powerful laptop for professionals
3	3	Sofa	2	SF789	299.99	10	Comfortable and stylish sofa
4	4	Dining Table	2	DT101	199.99	5	Wooden dining table with 4 chairs
5	5	T-shirt	3	TS111	19.99	200	Cotton t-shirt for casual wear
6	6	Jeans	3	JN222	39.99	150	Denim jeans for men
7	7	Pasta	4	FS333	2.99	500	Pack of dry pasta
8	8	Coffee	4	FB444	9.99	150	Ground coffee for brewing
9	9	Action Figure	5	TF555	12.99	100	Toy action figure for kids
10	10	Doll	5	TF666	15.99	80	Doll for young children
11	11	Fiction Novel	6	BK777	14.99	120	Mystery fiction novel
12	12	Cookbook	6	BK888	19.99	60	Recipes and cooking tips

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 12 rows

13.11. Suppliers Data

Suppliers Data.sql...F:\DUA LAPTOP (57) Tables.sql - DESKTOP...F:\DUA LAPTOP (55) Create&Use_Databa...DUA LAPTOP (53)

```
INSERT INTO Suppliers (Supplier_Name, Contact_Name, Phone_Number, Email, Address)
VALUES
('ABC Electronics', 'John Doe', '123-456-7890', 'contact@abc.com', '123 Tech Park, Silicon Valley'),
('XYZ Furniture', 'Jane Smith', '987-654-3210', 'support@xyzfurniture.com', '456 Wood St, Chicago'),
('Global Textiles', 'Sarah Johnson', '555-123-4567', 'service@globaltextiles.com', '789 Cotton Road, New York'),
('Gourmet Goods', 'Mike Brown', '666-789-1234', 'info@gourmetgoods.com', '101 Food Ave, LA'),
('Toy World', 'Emily Clark', '333-555-7777', 'sales@toyworld.com', '202 Toy Street, Dallas'),
('Book Haven', 'Alex White', '444-888-9999', 'contact@bookhaven.com', '303 Book Lane, Boston');

SELECT * FROM Suppliers
```

100 %

Results Messages

	Supplier_ID	Supplier_Name	Contact_Name	Phone_Number	Email	Address
1	1	ABC Electronics	John Doe	123-456-7890	contact@abc.com	123 Tech Park, Silicon Valley
2	2	XYZ Furniture	Jane Smith	987-654-3210	support@xyzfurniture.com	456 Wood St, Chicago
3	3	Global Textiles	Sarah Johnson	555-123-4567	service@globaltextiles.com	789 Cotton Road, New York
4	4	Gourmet Goods	Mike Brown	666-789-1234	info@gourmetgoods.com	101 Food Ave, LA
5	5	Toy World	Emily Clark	333-555-7777	sales@toyworld.com	202 Toy Street, Dallas
6	6	Book Haven	Alex White	444-888-9999	contact@bookhaven.com	303 Book Lane, Boston

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 6 rows

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13.12. Stock Transaction Data

Stock Transaction...F:\DUA LAPTOP (57) Tables.sql - DESKTOP...F:\DUA LAPTOP (55) Create&Use_Databa...DUA LAPTOP (53)

```
INSERT INTO Stock_Transactions (Product_ID, Transaction_Type, Quantity, Transaction_Date, Remarks)
VALUES
(1, 'Purchase', 30, '2025-01-01', 'Restocking new batch of smartphones'),
(2, 'Purchase', 15, '2025-01-02', 'New shipment of laptops'),
(3, 'Sale', 5, '2025-01-05', 'Customer purchased a sofa'),
(4, 'Sale', 3, '2025-01-06', 'Dining table sold to a customer'),
(5, 'Purchase', 200, '2025-01-01', 'New stock of t-shirts arrived'),
(6, 'Sale', 100, '2025-01-07', 'Jeans sold to customers'),
(7, 'Purchase', 500, '2025-01-03', 'Pasta stock replenished'),
(8, 'Sale', 150, '2025-01-08', 'Coffee sold to cafes'),
(9, 'Purchase', 100, '2025-01-04', 'Action figures added to inventory'),
(10, 'Sale', 80, '2025-01-05', 'Dolls purchased by customers'),
(11, 'Sale', 30, '2025-01-09', 'Fiction novel bought by customer'),
(12, 'Sale', 10, '2025-01-10', 'Cookbook bought by a customer');

SELECT * FROM Stock_Transactions
```

100 %

Results Messages

	Transaction_ID	Product_ID	Transaction_Type	Quantity	Transaction_Date	Remarks
1	1	1	Purchase	30	2025-01-01 00:00:00.000	Restocking new batch of smartphones
2	2	2	Purchase	15	2025-01-02 00:00:00.000	New shipment of laptops
3	3	3	Sale	5	2025-01-05 00:00:00.000	Customer purchased a sofa
4	4	4	Sale	3	2025-01-06 00:00:00.000	Dining table sold to a customer
5	5	5	Purchase	200	2025-01-01 00:00:00.000	New stock of t-shirts arrived
6	6	6	Sale	100	2025-01-07 00:00:00.000	Jeans sold to customers
7	7	7	Purchase	500	2025-01-03 00:00:00.000	Pasta stock replenished
8	8	8	Sale	150	2025-01-08 00:00:00.000	Coffee sold to cafes
9	9	9	Purchase	100	2025-01-04 00:00:00.000	Action figures added to inventory
10	10	10	Sale	80	2025-01-05 00:00:00.000	Dolls purchased by customers
11	11	11	Sale	30	2025-01-09 00:00:00.000	Fiction novel bought by customer

Query executed successfully. | DESKTOP-OCMKECF\SQLEXPRESS ... | DESKTOP-OCMKECF\DUA LA... | InventoryManagementSystem | 00:00:00 | 12 rows

13.13. Returns Data

Stock Transaction...F:\DUA LAPTOP (57) Tables.sql - DESKTOP...F:\DUA LAPTOP (55) Create&Use_Databa...DUA LAPTOP (53)

```
INSERT INTO Returns (Sale_Transaction_ID, Product_ID, Quantity, Reason, Return_Date)
VALUES
(3, 1, 1, 'Defective product', '2025-01-07'),
(4, 2, 1, 'Customer changed mind', '2025-01-08'),
(7, 5, 2, 'Not the right size', '2025-01-09'),
(8, 4, 3, 'Expired product', '2025-01-11'),
(9, 6, 1, 'Not what was expected', '2025-01-12');

SELECT * FROM Returns
```

100 %

Results Messages

	Return_ID	Sale_Transaction_ID	Product_ID	Quantity	Reason	Return_Date
1	1	3	1	1	Defective product	2025-01-07 00:00:00.000
2	2	4	2	1	Customer changed mind	2025-01-08 00:00:00.000
3	3	7	5	2	Not the right size	2025-01-09 00:00:00.000
4	4	8	4	3	Expired product	2025-01-11 00:00:00.000
5	5	9	6	1	Not what was expected	2025-01-12 00:00:00.000

Query executed successfully. | DESKTOP-OCMKECF\SQLEXPRESS ... | DESKTOP-OCMKECF\DUA LA... | InventoryManagementSystem | 00:00:00 | 5 rows

INVENTORY MANAGEMENT SYSTEM

13.14. Customer Data

Stock Transaction...F:\DUA LAPTOP (57) Tables.sql - DESKTOP...F:\DUA LAPTOP (55)* X Create&Use_Databa...DUA LAPTOP (53)

```
INSERT INTO Customers (Customer_Name, Phone_Number, Email, Address)
VALUES
('Alice Johnson', '555-0101', 'alice.johnson@example.com', '101 Elm St, Chicago'),
('Bob Williams', '555-0202', 'bob.williams@example.com', '202 Oak St, New York'),
('Charlie Brown', '555-0303', 'charlie.brown@example.com', '303 Pine St, LA'),
('David White', '555-0404', 'david.white@example.com', '404 Maple St, Boston'),
('Eve Green', '555-0505', 'eve.green@example.com', '505 Birch St, Dallas'),
('Frank Black', '555-0606', 'frank.black@example.com', '606 Cedar St, San Francisco'),
('Grace Adams', '555-0707', 'grace.adams@example.com', '707 Willow St, Miami'),
('Helen Clark', '555-0808', 'helen.clark@example.com', '808 Redwood St, Seattle'),
('Ivy King', '555-0909', 'ivy.king@example.com', '909 Ash St, Phoenix'),
('Jack Lewis', '555-1010', 'jack.lewis@example.com', '1010 Pine St, Denver');

SELECT * FROM Customers
```

100 %

Results Messages

	Customer_ID	Customer_Name	Phone_Number	Email	Address
1	1	Alice Johnson	555-0101	alice.johnson@example.com	101 Elm St, Chicago
2	2	Bob Williams	555-0202	bob.williams@example.com	202 Oak St, New York
3	3	Charlie Brown	555-0303	charlie.brown@example.com	303 Pine St, LA
4	4	David White	555-0404	david.white@example.com	404 Maple St, Boston
5	5	Eve Green	555-0505	eve.green@example.com	505 Birch St, Dallas
6	6	Frank Black	555-0606	frank.black@example.com	606 Cedar St, San Francisco
7	7	Grace Adams	555-0707	grace.adams@example.com	707 Willow St, Miami
8	8	Helen Clark	555-0808	helen.clark@example.com	808 Redwood St, Seattle
9	9	Ivy King	555-0909	ivy.king@example.com	909 Ash St, Phoenix
10	10	Jack Lewis	555-1010	jack.lewis@example.com	1010 Pine St, Denver

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 10 rows

13.15. Purchase Order Data

Stock Transaction...F:\DUA LAPTOP (57) Tables.sql - DESKTOP...F:\DUA LAPTOP (55)* X Create&Use_Databa...DUA LAPTOP (53)

```
INSERT INTO Purchase_Orders (Supplier_ID, Order_Date, Expected_Delivery_Date, Status, Total_Amount)
VALUES
(1, '2025-01-05', '2025-01-10', 'Pending', 14999.70),
(2, '2025-01-06', '2025-01-12', 'Shipped', 2999.70),
(3, '2025-01-07', '2025-01-13', 'Delivered', 7999.50),
(4, '2025-01-02', '2025-01-08', 'Delivered', 2999.10),
(5, '2025-01-04', '2025-01-11', 'Pending', 1559.90),
(6, '2025-01-03', '2025-01-09', 'Shipped', 1799.70);

SELECT * FROM Purchase_Orders
```

100 %

Results Messages

	Order_ID	Supplier_ID	Order_Date	Expected_Delivery_Date	Status	Total_Amount
1	1	1	2025-01-05 00:00:00.000	2025-01-10 00:00:00.000	Pending	14999.70
2	2	2	2025-01-06 00:00:00.000	2025-01-12 00:00:00.000	Shipped	2999.70
3	3	3	2025-01-07 00:00:00.000	2025-01-13 00:00:00.000	Delivered	7999.50
4	4	4	2025-01-02 00:00:00.000	2025-01-08 00:00:00.000	Delivered	2999.10
5	5	5	2025-01-04 00:00:00.000	2025-01-11 00:00:00.000	Pending	1559.90
6	6	6	2025-01-03 00:00:00.000	2025-01-09 00:00:00.000	Shipped	1799.70

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 6 rows

INVENTORY MANAGEMENT SYSTEM

13.16. Total Sales

The screenshot shows a SQL Server Enterprise Manager window with a query executed. The query is:

```
SELECT SUM(Quantity) AS Total_Sales
FROM Stock_Transactions
WHERE Transaction_Type = 'Sale';
```

The results pane shows a single row with the value 378 for the column Total_Sales.

	Total_Sales
1	378

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 1 rows

13.17. Average Price

The screenshot shows a SQL Server Enterprise Manager window with a query executed. The query is:

```
SELECT AVG(Price) AS Average_Price
FROM Products;
```

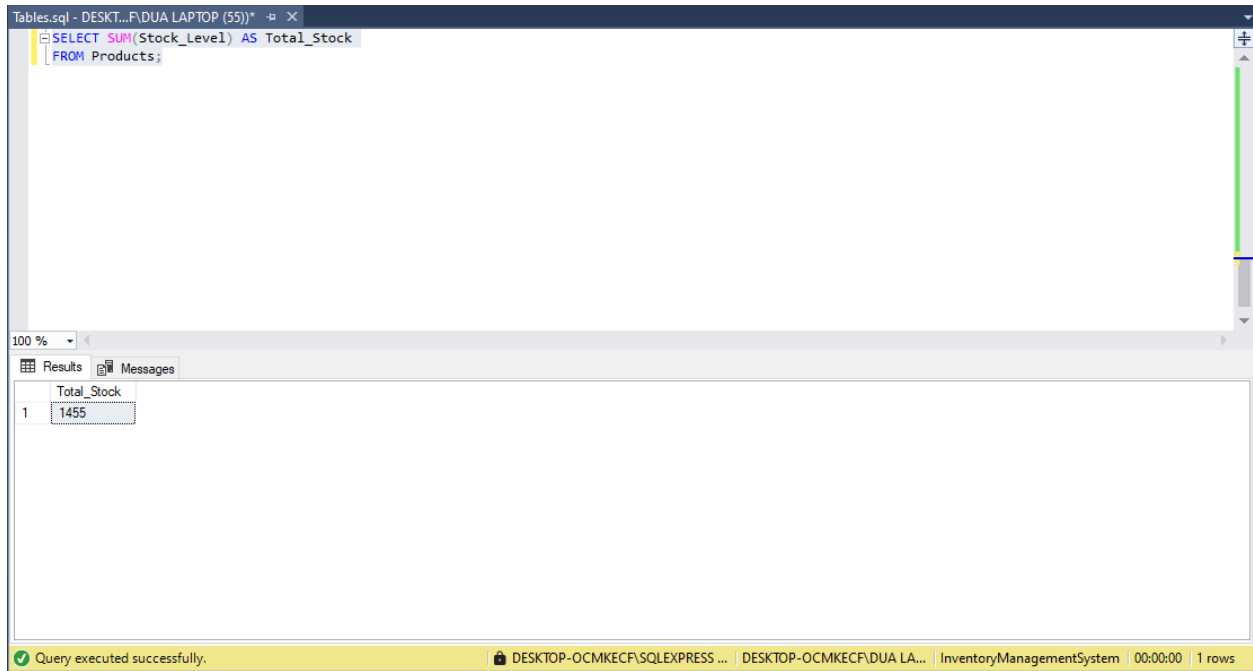
The results pane shows a single row with the value 169.740000 for the column Average_Price.

	Average_Price
1	169.740000

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 1 rows

INVENTORY MANAGEMENT SYSTEM

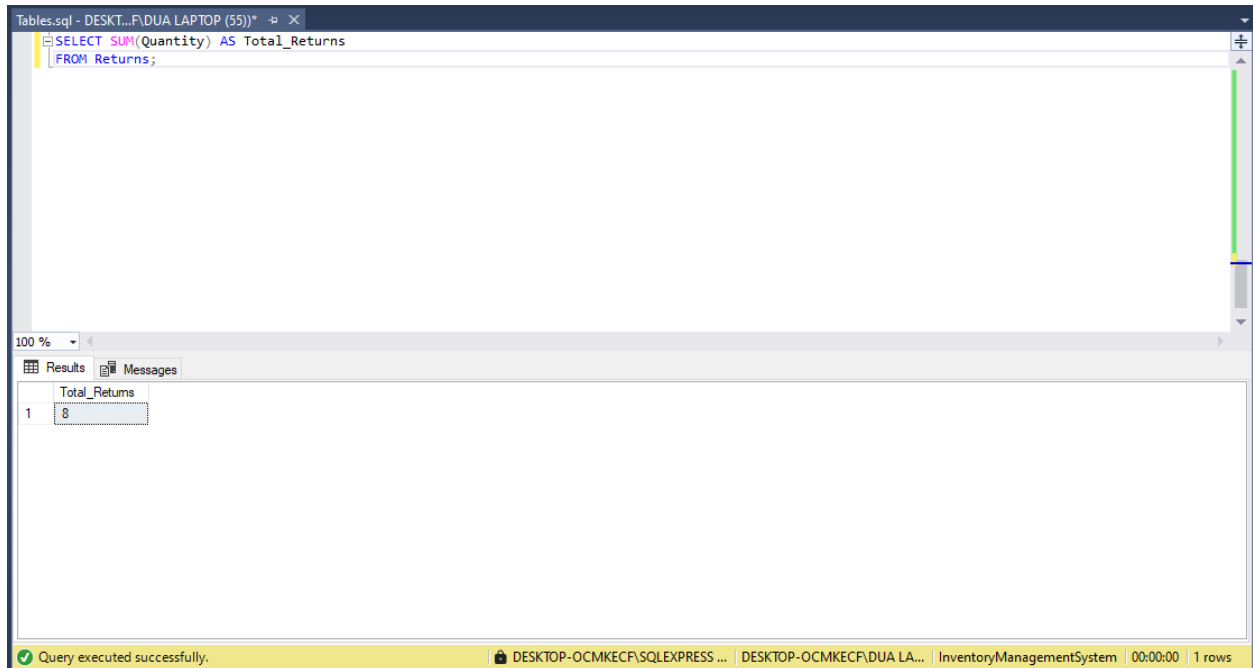
13.18. Total Stocks



The screenshot shows a SQL Server Enterprise Manager window with a query executed. The query is: `SELECT SUM(Stock_Level) AS Total_Stock FROM Products;`. The results pane shows a single row with the value 1455 for the column Total_Stock. The status bar at the bottom indicates the query was executed successfully, returning 1 row.

	Total_Stock
1	1455

13.19. Total Returns



The screenshot shows a SQL Server Enterprise Manager window with a query executed. The query is: `SELECT SUM(Quantity) AS Total_Returns FROM Returns;`. The results pane shows a single row with the value 8 for the column Total_Returns. The status bar at the bottom indicates the query was executed successfully, returning 1 row.

	Total_Returns
1	8

INVENTORY MANAGEMENT SYSTEM

13.20. Max Quantity Sold

The screenshot shows a SQL query window with the following text:

```
SELECT MAX(Quantity) AS Max_Quantity_Sold
FROM Stock_Transactions
WHERE Transaction_Type = 'Sale';
```

Below the query window, the 'Results' tab is active, displaying a single row of data:

	Max_Quantity_Sold
1	150

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 1 rows

13.21. Total Amount for Purchase Orders

The screenshot shows a SQL query window with the following text:

```
SELECT SUM(Total_Amount) AS Total_Amount_For_Purchase_Orders
FROM Purchase_Orders;
```

Below the query window, the 'Results' tab is active, displaying a single row of data:

	Total_Amount_For_Purchase_Orders
1	32357.60

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 1 rows

INVENTORY MANAGEMENT SYSTEM

13.22. Number of Products Sold by Category

Tables.sql - DESKTOP...F\DUALAPTOP (55)

```
SELECT p.Category_ID, COUNT(st.Product_ID) AS Number_of_Products_Sold
FROM Stock_Transactions st
JOIN Products p ON st.Product_ID = p.Product_ID
WHERE st.Transaction_Type = 'Sale'
GROUP BY p.Category_ID;
```

100 %

Results Messages

	Category_ID	Number_of_Products_Sold
1	2	2
2	3	1
3	4	1
4	5	1
5	6	2

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUALAPTOP (55) InventoryManagementSystem 00:00:00 5 rows

13.23. Inner Join: Get product details along with their category information

Tables.sql - DESKTOP...F\DUALAPTOP (55)

```
SELECT p.Product_ID, p.Product_Name, p.SKU, p.Price, c.Category_Name
FROM Products p
JOIN Categories c ON p.Category_ID = c.Category_ID;
```

100 %

Results Messages

	Product_ID	Product_Name	SKU	Price	Category_Name
1	1	Smartphone	SP123	499.99	Electronics
2	2	Laptop	LT456	899.99	Electronics
3	3	Sofa	SF789	299.99	Furniture
4	4	Dining Table	DT101	199.99	Furniture
5	5	T-shirt	TS111	19.99	Clothing
6	6	Jeans	JN222	39.99	Clothing
7	7	Pasta	FS333	2.99	Food & Beverages
8	8	Coffee	FB444	9.99	Food & Beverages
9	9	Action Figure	TF555	12.99	Toys
10	10	Doll	TF666	15.99	Toys
11	11	Fiction Novel	BK777	14.99	Books
12	12	Cookbook	BK888	19.99	Books

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUALAPTOP (55) InventoryManagementSystem 00:00:00 12 rows

INVENTORY MANAGEMENT SYSTEM

13.24. Left Join: Get all products along with their category details

Tables.sql - DESKTOP...F.DUA LAPTOP (55)

```
SELECT p.Product_ID, p.Product_Name, p.SKU, p.Price, c.Category_Name
FROM Products p
LEFT JOIN Categories c ON p.Category_ID = c.Category_ID;
```

100 %

Results Messages

	Product_ID	Product_Name	SKU	Price	Category_Name
1	1	Smartphone	SP123	499.99	Electronics
2	2	Laptop	LT456	899.99	Electronics
3	3	Sofa	SF789	299.99	Furniture
4	4	Dining Table	DT101	199.99	Furniture
5	5	T-shirt	TS111	19.99	Clothing
6	6	Jeans	JN222	39.99	Clothing
7	7	Pasta	FS333	2.99	Food & Beverages
8	8	Coffee	FB444	9.99	Food & Beverages
9	9	Action Figure	TF555	12.99	Toys
10	10	Doll	TF666	15.99	Toys
11	11	Fiction Novel	BK777	14.99	Books
12	12	Cookbook	BK888	19.99	Books

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 12 rows

13.25. Right Join: Get all categories and their products

Tables.sql - DESKTOP...F.DUA LAPTOP (55)

```
SELECT c.Category_ID, c.Category_Name, p.Product_ID, p.Product_Name
FROM Categories c
RIGHT JOIN Products p ON p.Category_ID = c.Category_ID;
```

100 %

Results Messages

	Category_ID	Category_Name	Product_ID	Product_Name
1	1	Electronics	1	Smartphone
2	1	Electronics	2	Laptop
3	2	Furniture	3	Sofa
4	2	Furniture	4	Dining Table
5	3	Clothing	5	T-shirt
6	3	Clothing	6	Jeans
7	4	Food & Beverages	7	Pasta
8	4	Food & Beverages	8	Coffee
9	5	Toys	9	Action Figure
10	5	Toys	10	Doll
11	6	Books	11	Fiction Novel
12	6	Books	12	Cookbook

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 12 rows

INVENTORY MANAGEMENT SYSTEM

13.26. Full Join: Get all products and categories, whether they are related or not

The screenshot shows a SQL query window with the following query:

```
SELECT p.Product_ID, p.Product_Name, c.Category_Name
FROM Products p
FULL JOIN Categories c ON p.Category_ID = c.Category_ID;
```

The results pane displays 12 rows of data:

Product_ID	Product_Name	Category_Name
1	Smartphone	Electronics
2	Laptop	Electronics
3	Sofa	Furniture
4	Dining Table	Furniture
5	T-shirt	Clothing
6	Jeans	Clothing
7	Pasta	Food & Beverages
8	Coffee	Food & Beverages
9	Action Figure	Toys
10	Doll	Toys
11	Fiction Novel	Books
12	Cookbook	Books

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 12 rows

13.27. Sub Query: Total Sales for a Product

The screenshot shows a SQL query window with the following query:

```
SELECT Product_Name,
(SELECT SUM(Quantity)
FROM Stock_Transactions
WHERE Product_ID = p.Product_ID AND Transaction_Type = 'Sale') AS Total_Sales
FROM Products p;
```

The results pane displays 12 rows of data:

Product_Name	Total_Sales
Smartphone	NULL
Laptop	NULL
Sofa	5
Dining Table	3
T-shirt	NULL
Jeans	100
Pasta	NULL
Coffee	150
Action Figure	NULL
Doll	80
Fiction Novel	30
Cookbook	10

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 12 rows

INVENTORY MANAGEMENT SYSTEM

13.28. Sub Query: Average Price for a Category

The screenshot shows a SQL query window with the following query:

```
SELECT Category_Name,
       (SELECT AVG(Price)
        FROM Products
        WHERE Category_ID = c.Category_ID) AS Average_Price
FROM Categories c;
```

The query results are displayed in a table with 6 rows:

	Category_Name	Average_Price
1	Electronics	699.990000
2	Furniture	249.990000
3	Clothing	29.990000
4	Food & Beverages	6.490000
5	Toys	14.490000
6	Books	17.490000

The status bar at the bottom indicates: Query executed successfully. | DESKTOP-OCMKECF\SQLEXPRESS ... | DESKTOP-OCMKECF\DUA LA... | InventoryManagementSystem | 00:00:00 | 6 rows

13.29. Sub Query: Total Stock for a Product

The screenshot shows a SQL query window with the following query:

```
SELECT Product_Name,
       (SELECT SUM(Stock_Level)
        FROM Products
        WHERE Product_ID = p.Product_ID) AS Total_Stock
FROM Products p;
```

The query results are displayed in a table with 12 rows:

	Product_Name	Total_Stock
1	Smartphone	50
2	Laptop	30
3	Sofa	10
4	Dining Table	5
5	T-shirt	200
6	Jeans	150
7	Pasta	500
8	Coffee	150
9	Action Figure	100
10	Doll	80
11	Fiction Novel	120
12	Cookbook	60

The status bar at the bottom indicates: Query executed successfully. | DESKTOP-OCMKECF\SQLEXPRESS ... | DESKTOP-OCMKECF\DUA LA... | InventoryManagementSystem | 00:00:00 | 12 rows

INVENTORY MANAGEMENT SYSTEM

13.30. Nested Query: Find Products with Price Greater than the Average Price

The screenshot shows a SQL query window with the following query:

```
SELECT Product_Name, Price
FROM Products p
WHERE p.Price > (
    SELECT AVG(Price)
    FROM Products
);
```

The query results are displayed in a table with the following data:

	Product_Name	Price
1	Smartphone	499.99
2	Laptop	899.99
3	Sofa	299.99
4	Dining Table	199.99

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 4 rows

13.31. Nested Query: Find Products in a Specific Category

The screenshot shows a SQL query window with the following query:

```
SELECT Product_Name
FROM Products p
WHERE p.Category_ID = (
    SELECT Category_ID
    FROM Categories
    WHERE Category_Name = 'Electronics'
);
```

The query results are displayed in a table with the following data:

	Product_Name
1	Smartphone
2	Laptop

The status bar at the bottom indicates: Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 2 rows

INVENTORY MANAGEMENT SYSTEM

13.32. View: Stock Levels with Product Details

Tables.sql - DESKTOP...F:\DUA LAPTOP (55))

```
CREATE VIEW Stock_Levels_With_Details AS
SELECT
    p.Product_ID,
    p.Product_Name,
    p.SKU,
    p.Category_ID,
    p.Stock_Level,
    p.Price,
    p.Description
FROM
    Products p;

SELECT * FROM Stock_Levels_With_Details
```

Results

	Product_ID	Product_Name	SKU	Category_ID	Stock_Level	Price	Description
1	1	Smartphone	SP123	1	50	499.99	Latest model with high-end features
2	2	Laptop	LT456	1	30	899.99	Powerful laptop for professionals
3	3	Sofa	SF789	2	10	299.99	Comfortable and stylish sofa
4	4	Dining Table	DT101	2	5	199.99	Wooden dining table with 4 chairs
5	5	T-shirt	TS111	3	200	19.99	Cotton t-shirt for casual wear
6	6	Jeans	JN222	3	150	39.99	Denim jeans for men
7	7	Pasta	FS333	4	500	2.99	Pack of dry pasta
8	8	Coffee	FB444	4	150	9.99	Ground coffee for brewing
9	9	Action Figure	TF555	5	100	12.99	Toy action figure for kids
10	10	Doll	TF666	5	80	15.99	Doll for young children
11	11	Fiction Novel	BK777	6	120	14.99	Mystery fiction novel
12	12	Cookbook	BK888	6	60	19.99	Recipes and cooking tips

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 12 rows

13.33. View: Product Returns Summary

Tables.sql - DESKTOP...F:\DUA LAPTOP (55))

```
CREATE VIEW Product_Returns_Summary AS
SELECT
    p.Product_ID,
    p.Product_Name,
    SUM(r.Quantity) AS Total_Returned_Quantity,
    STRING_AGG(r.Reason, ', ') AS Reasons
FROM
    Products p
JOIN
    Returns r ON p.Product_ID = r.Product_ID
GROUP BY
    p.Product_ID, p.Product_Name;

SELECT * FROM Product_Returns_Summary
```

Results

	Product_ID	Product_Name	Total_Returned_Quantity	Reasons
1	1	Smartphone	1	Defective product
2	2	Laptop	1	Customer changed mind
3	4	Dining Table	3	Expired product
4	5	T-shirt	2	Not the right size
5	6	Jeans	1	Not what was expected

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 5 rows

INVENTORY MANAGEMENT SYSTEM

13.34. Stored Procedure with Transaction and Rollback: Add Stock to a Product

```
SQLQuery2.sql - D:\F\DUA LAPTOP (62))  SQLQuery1.sql - D:\F\DUA LAPTOP (53)) * X
CREATE PROC Add_Stock
    @Product_ID INT,
    @Quantity INT,
    @Transaction_Date DATETIME,
    @Remarks NVARCHAR(500)
AS
BEGIN
    BEGIN TRANSACTION;

    BEGIN TRY
        UPDATE Products
        SET Stock_Level = Stock_Level + @Quantity
        WHERE Product_ID = @Product_ID;

        INSERT INTO Stock_Transactions (Product_ID, Transaction_Type, Quantity, Transaction_Date, Remarks)
        VALUES (@Product_ID, 'Restock', @Quantity, @Transaction_Date, @Remarks);

        COMMIT TRANSACTION;
        PRINT 'Stock added successfully.';
    END TRY

    BEGIN CATCH
        ROLLBACK TRANSACTION;
        PRINT 'Error adding stock. Rolling back.';
        THROW;
    END CATCH;
END;
```

100 % Query executed successfully. DESKTOP-OCMKECF/SQLEXPRESS ... DESKTOP-OCMKECF/DUA LA... InventoryManagementSystem 00:00:00 12 rows

Before Updating Dinning Table Stock:

	Product_ID	Product_Name	Category_ID	SKU	Price	Stock_Level	Description
1	1	Smartphone	1	SP123	499.99	50	Latest model with high-end features
2	2	Laptop	1	LT456	899.99	30	Powerful laptop for professionals
3	3	Sofa	2	SF789	299.99	10	Comfortable and stylish sofa
4	4	Dining Table	2	DT101	199.99	5	Wooden dining table with 4 chairs
5	5	T-shirt	3	TS111	19.99	200	Cotton t-shirt for casual wear
6	6	Jeans	3	JN222	39.99	150	Denim jeans for men
7	7	Pasta	4	FS333	2.99	500	Pack of dry pasta
8	8	Coffee	4	FB444	9.99	150	Ground coffee for brewing
9	9	Action Figure	5	TF555	12.99	100	Toy action figure for kids
10	10	Doll	5	TF666	15.99	80	Doll for young children
11	11	Fiction Novel	6	BK777	14.99	120	Mystery fiction novel
12	12	Cookbook	6	BK888	19.99	60	Recipes and cooking tips

INVENTORY MANAGEMENT SYSTEM

After Updating Dinnig Table Stock:

The screenshot shows the SQL Server Enterprise Manager interface. The top pane displays the executed query: `EXEC Add_Stock 4, 50, '2025-1-09', 'Updated Dinning Table Stock';`. The bottom pane, titled 'Messages', shows the following output:
(1 row affected)
(1 row affected)
Stock added successfully.
Completion time: 2025-01-09T16:13:40.4811377+05:00
The status bar at the bottom indicates 'Query executed successfully.' and shows the server name 'DESKTOP-OCMKECF\SQLEXPRESS' and the database 'InventoryManagementSystem'.

Updated Stock:

The screenshot shows the SQL Server Enterprise Manager interface. The top pane displays the query: `SELECT * FROM Products`. The bottom pane, titled 'Results', shows the following table of data:

	Product_ID	Product_Name	Category_ID	SKU	Price	Stock_Level	Description
1	1	Smartphone	1	SP123	499.99	50	Latest model with high-end features
2	2	Laptop	1	LT456	899.99	30	Powerful laptop for professionals
3	3	Sofa	2	SF789	299.99	10	Comfortable and stylish sofa
4	4	Dining Table	2	DT101	199.99	55	Wooden dining table with 4 chairs
5	5	T-shirt	3	TS111	19.99	200	Cotton t-shirt for casual wear
6	6	Jeans	3	JN222	39.99	150	Denim jeans for men
7	7	Pasta	4	FS333	2.99	500	Pack of dry pasta
8	8	Coffee	4	FB444	9.99	150	Ground coffee for brewing
9	9	Action Figure	5	TF555	12.99	100	Toy action figure for kids
10	10	Doll	5	TF666	15.99	80	Doll for young children
11	11	Fiction Novel	6	BK777	14.99	120	Mystery fiction novel
12	12	Cookbook	6	BK888	19.99	60	Recipes and cooking tips

The status bar at the bottom indicates 'Query executed successfully.' and shows the server name 'DESKTOP-OCMKECF\SQLEXPRESS' and the database 'InventoryManagementSystem'.

Inventory Management System

13.35. Stored Procedure with Transaction and Rollback: Update Product Price

```
SQLQuery1.sql - D:\DUA LAPTOP (53)) * X
CREATE PROC Update_Product_Price
    @Product_ID INT,
    @New_Price DECIMAL(10, 2)
AS
DECLARE @ProductExists INT;
BEGIN
    BEGIN TRANSACTION;

    SELECT @ProductExists = COUNT(1)
    FROM Products
    WHERE Product_ID = @Product_ID;

    IF @ProductExists = 0
    BEGIN
        ROLLBACK TRANSACTION;
        PRINT 'Product does not exist. Transaction rolled back.';
        RETURN;
    END

    UPDATE Products
    SET Price = @New_Price
    WHERE Product_ID = @Product_ID;

    COMMIT TRANSACTION;
    PRINT 'Product price updated successfully.';
END;
```

100 %
Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 0 rows

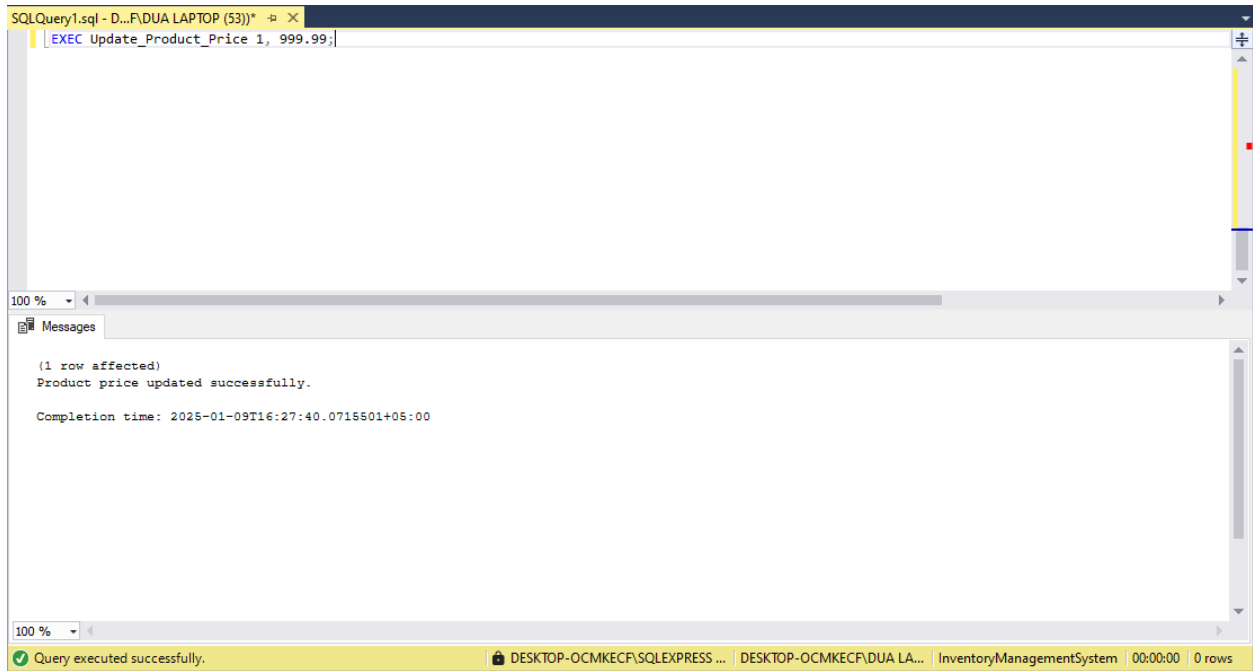
Before Updating Price Smartphone Price:

	Product_ID	Product_Name	Category_ID	SKU	Price	Stock_Level	Description
1	1	Smartphone	1	SP123	499.99	50	Latest model with high-end features
2	2	Laptop	1	LT456	899.99	30	Powerful laptop for professionals
3	3	Sofa	2	SF789	299.99	10	Comfortable and stylish sofa
4	4	Dining Table	2	DT101	199.99	55	Wooden dining table with 4 chairs
5	5	T-shirt	3	TS111	19.99	200	Cotton t-shirt for casual wear
6	6	Jeans	3	JN222	39.99	150	Denim jeans for men
7	7	Pasta	4	FS333	2.99	500	Pack of dry pasta
8	8	Coffee	4	FB444	9.99	150	Ground coffee for brewing
9	9	Action Figure	5	TF555	12.99	100	Toy action figure for kids
10	10	Doll	5	TF666	15.99	80	Doll for young children
11	11	Fiction Novel	6	BK777	14.99	120	Mystery fiction novel
12	12	Cookbook	6	BK888	19.99	60	Recipes and cooking tips

Query executed successfully. DESKTOP-OCMKECF\SQLEXPRESS ... DESKTOP-OCMKECF\DUA LA... InventoryManagementSystem 00:00:00 12 rows

Inventory Management System

After Updating Smartphone Price:



Updated Price:

Results

Messages

	Product_ID	Product_Name	Category_ID	SKU	Price	Stock_Level	Description
1	1	Smartphone	1	SP123	999.99	50	Latest model with high-end features
2	2	Laptop	1	LT456	899.99	30	Powerful laptop for professionals
3	3	Sofa	2	SF789	299.99	10	Comfortable and stylish sofa
4	4	Dining Table	2	DT101	199.99	55	Wooden dining table with 4 chairs
5	5	T-shirt	3	TS111	19.99	200	Cotton t-shirt for casual wear
6	6	Jeans	3	JN222	39.99	150	Denim jeans for men
7	7	Pasta	4	FS333	2.99	500	Pack of dry pasta
8	8	Coffee	4	FB444	9.99	150	Ground coffee for brewing
9	9	Action Figure	5	TF555	12.99	100	Toy action figure for kids
10	10	Doll	5	TF666	15.99	80	Doll for young children
11	11	Fiction Novel	6	BK777	14.99	120	Mystery fiction novel
12	12	Cookbook	6	BK888	19.99	60	Recipes and cooking tips

Query executed successfully.

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12 rows

14. Backup:

- [InventoryManagementSystem](#)